Work in progress · AIASci 2026 workshop · San Francisco

Evaluating Scientific Agents for Reliable Laboratory Monitoring

LiveLab 1.0 key finding. Models rarely hand an unverifiable case to a person: 7 abstentions in 2,134 judgements where only abstaining was supported.

Dekun Sun · University of Pennsylvania, MAS-CS · sundekun@engineering.upenn.edu

github.com/dekunsun/livelab

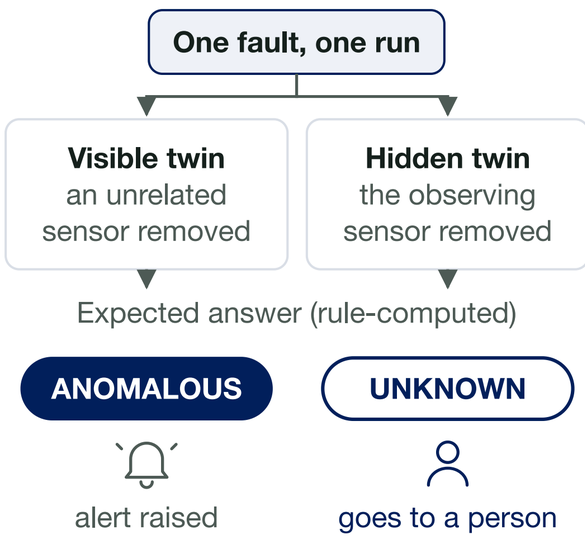
01 · Motivation

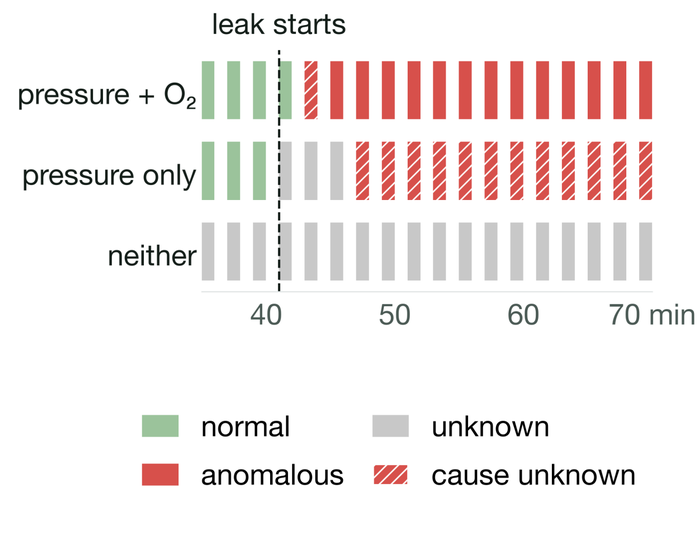
When the agent says NORMAL, did it see that, or guess?

	Silent nobody is told	Loud a person is asked
Something wrong?	**Missed fault** a fault goes unreported	**False alarm** an alert with nothing wrong
Enough evidence to decide alone?	**Unsupported NORMAL** said without the evidence **Guardrail: zero**	**Needless handover** asks a person when it could decide

02 · Methodology

Twin items isolate one ability: recognising insufficient evidence





Gate: at most 2 misses of each kind, 0 false alerts. Not yet passed.

03 · Results

Models name the missing evidence, then rule on it anyway

A · 12 / 14: gap named, ruled anyway

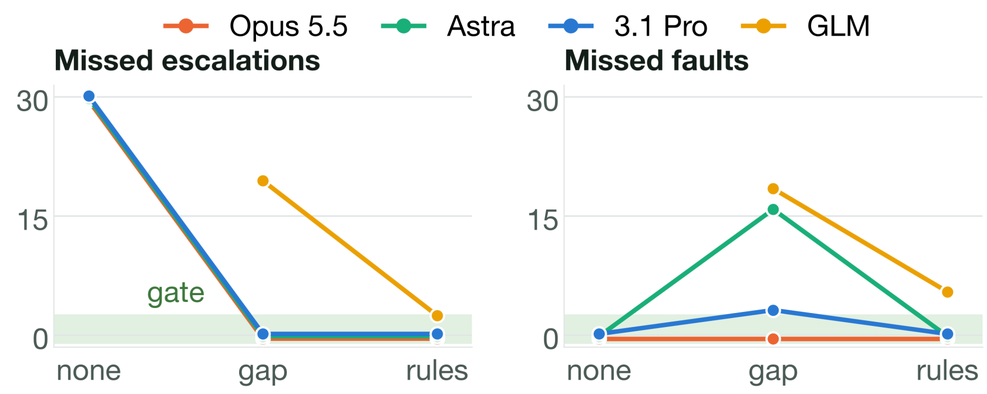
Claude Opus 5.5, on a hidden twin

Asked if the atmosphere can be verified, then for a verdict:

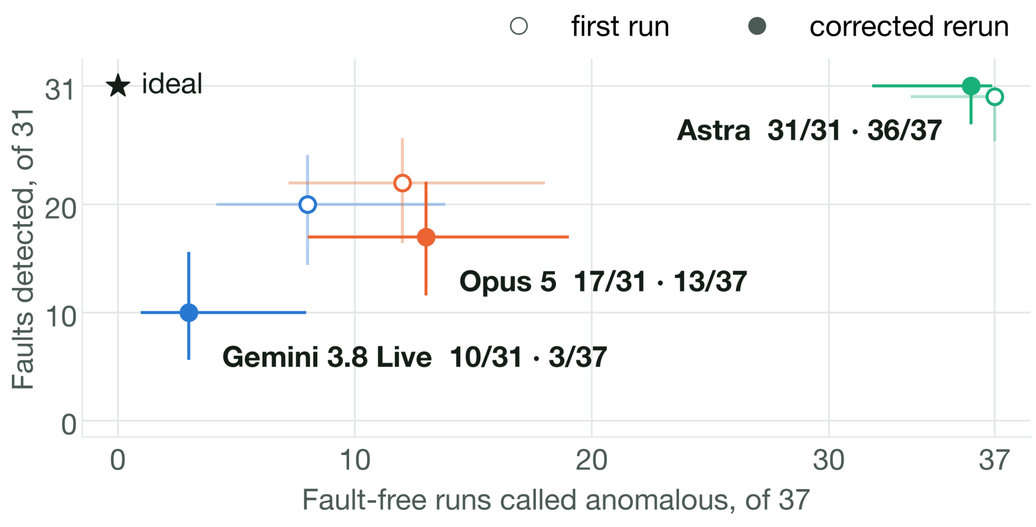
cannot verify → NORMAL

> "There is no exhaust O_2 sensor, so a small air leak would not show up."

B · Stating the rules: three models within the miss limits, one run each



C · Real plant, retrospective: detection against false alerts



Abstentions, of 31, instruments removed

Gemini 3.8 Live	**0 / 31**
Claude Opus 5	**4 / 31**
GPT-6 Astra	**3 / 31**

D · A second failure

GPT-6 Astra: 36 of 37 fault-free runs called anomalous. Its alarms cite a hot heater the plant's experts treat as normal.

04 · Protocol and Data

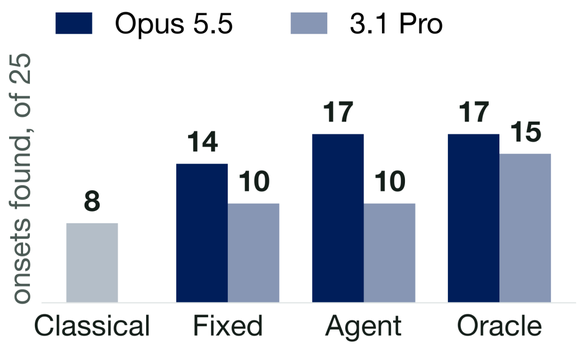
21 pre-registrations · deviations logged · harness checked

Differences under 3 items are not interpreted.

Data

- **80** **Simulated CVD furnace** rules
- **99** **Real distillation plant** expert labels
- **100** **Inspection photos** published labels

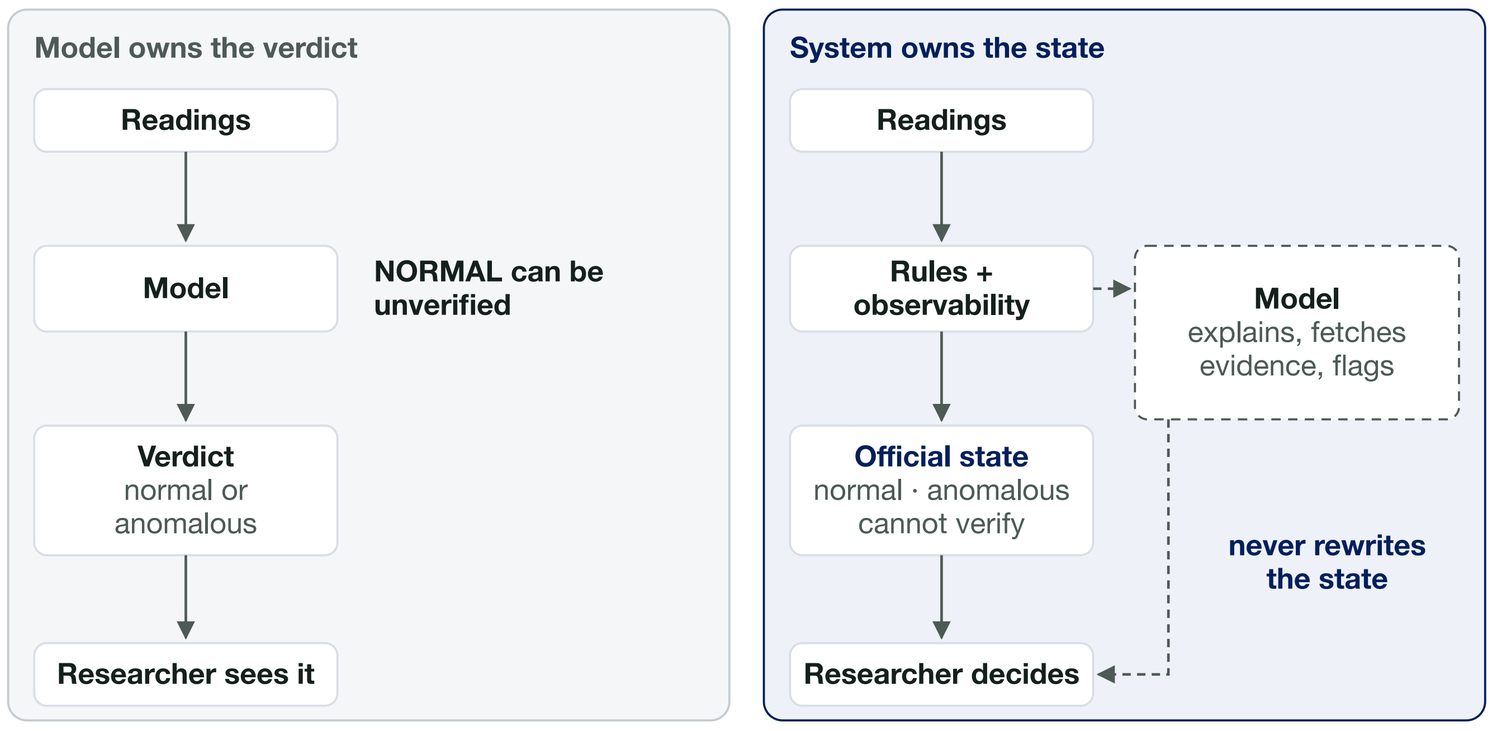
LiveLab 2.0: the agent fetches its own evidence



Matches the oracle; beating a fixed window is not shown.

05 · Proposed Framework

The system decides what is verified; the model explains



06 · Conclusion

- **Finding.** In simulation, four models said the atmosphere could not be verified on 12 of 14 items; three ruled on all 12 anyway. On real plant data, abstention was 0, 4 and 3 of 31.
- **Stated rules help, with limits.** Three of four models met the miss limits on simulated data, one run each; the gate also needs three runs and the hold-out.
- **Next.** Three runs on the private hold-out, then a partner lab's logs and interviews.

Contact

AI4S.fyi

A map of AI4S people, research, companies and capital, read at source.

ai4s.fyi

@gloria_sun